03

高分子化学

自由基聚合

内容大纲

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4. 关于动力学方程的讨论

- 使用条件和范围
  - 等活性理论
  - 聚合度很大
  - 自由基稳态

$$R = k_p \left(\frac{f k_d}{k_t} \right)^{\frac{1}{2}} [M][I]^{\frac{1}{2}}$$

  - 无单体链转移
  - 双基终止
  - 低转化率
  - 引发剂引发且R_d为控速步骤

- $R_p \propto [\,I\,]^n$

$$R = k_p[M\cdot][M]$$

自由基稳态假设 $\begin{cases} R_i \\ R_t \end{cases}$

$$R_p = k_p[M][M\cdot]$$

低转化率

自由基稳态 $R_i = R_t$

双基终止 $R_t = 2k_t[M\cdot]^2$

$[M\cdot] = (R_i/2k_t)^{0.5}$

$$= k_p[M](R_i/2k_t)^{0.5}$$

$$= k_p[M](R_d/2k_t)^{0.5}$$

引发剂引发且分解反应为控速步

$R_i = R_d = 2fk_d[I]$

$$= k_p\,(fk_d/k_t)^{0.5}\,[M][I]^{0.5}$$

引发剂引发

双基终止

$$R_p \propto [I]^{0.5}$$

$R_p = k_p[M][M\cdot]$

低转化率

自由基稳态 $R_i = R_t$

双基终止 $R_t = 2k_t[M\cdot]^2$

$[M\cdot] = (R_i/2k_t)^{0.5}$

$= k_p[M](R_i/2k_t)^{0.5}$

$= k_p\,(k_i/2k_t)^{0.5}\,[M]^{2.5}$

热引发

$R_i = k_i[M]^3$

热引发

双基终止

$R_p \propto [I]^0$

$$R_p = k_p[M][M\cdot]$$

$$M\bullet + R'H \xrightarrow{k_t} MH + R'\bullet$$

(无活性)

$$= k_p[M]R_i/k_t'$$

$$= k_p[M]R_d/k_t$$

$$= k_p(2fk_d/k_t)[M][I]^1$$

低转化率

自由基稳态 $R_i = R_t$

单基终止

$R_t = k_t[M\cdot][RH] = k_t'[M\cdot]$

$[M\cdot] = R_t/k_t' = R_i/k_t'$

引发剂引发且分解反应为控速步

$R_i = R_d = 2fk_d[I]$

引发剂引发

单基终止

$R_p \propto [I]^1$

- $R_p \propto [M]^m$

$$R = k_p[M\cdot][M]$$

自由基稳态假设

$$\begin{cases} R_i = R_d = 2fk_d[I] \\ R_t = 2k_t\left[\mathrm{M}_n^{\bullet}\right]^2 \end{cases}$$

$$\boxed{R = k_p\left(\frac{fk_d}{k_t}\right)^{\frac{1}{2}}[M][I]^{\frac{1}{2}}}$$

$$R_p = k[M]^{1\sim2}[I]^{0\sim1}$$

例1

St，甲苯溶剂，BPO，80℃

[M]=0.4mol/L m=1.36 ?

引发剂引发($k_i > k_d$)

$R_i = 2fk_d[M]^x[I]$, x = 0~1

$$R_p = k[M]^{1\sim2}[I]^{0\sim1}$$

例2

氯乙烯,本体聚合 n= 0.5-0.6

例3

丙烯腈,本体聚合，n= 0.9

氯乙烯、丙烯腈与其聚合物不相溶，链自由基末端被包围，出现单基终止与双基终止并存的现象

5. 自由基寿命

- 稳态假设成立

$$\tau = \frac{\text{自由基总浓度}}{\text{自由基产生（或消耗）速度}} = \frac{[\mathrm{M}\cdot]}{2fk_d[\mathrm{I}]} = \frac{\cancel{[\mathrm{M}\cdot]}}{2k_t[\mathrm{M}\cdot]^{\cancel{2}}}$$

$$= \frac{k_p[M]}{2k_tR_p} = \frac{1}{2\left(fk_tk_d[\mathrm{I}]\right)^{1/2}} = \frac{1}{(2k_tR_i)^{1/2}}$$

$$[M\cdot] = \left(\frac{fk_d[I]}{k_t}\right)^{\frac{1}{2}}$$

➢ 若取 $f=1$，$k_d=10^{-5}\mathrm{s}^{-1}$，$k_t=10^{7}\ \mathrm{L\cdot mol^{-1}\cdot s^{-1}}$，$[\mathrm{I}]=10^{-3}\ \mathrm{mol\cdot L^{-1}}$

得 $\tau \approx 1.58$ s

➢ τ：$10^{-1} \sim 10^{1}$ s

3.5 自由基聚合的相对分子量（聚合度）

- 影响聚合度大小的主要因素

引发剂浓度

单体浓度

温度

终止方式

链转移

1. 动力学链长 (Kinetics Chain Length)(ν)

$\bar{\nu}$

平均每一个活性中心从引发开始到真正消失这一过程中所消耗的单体分子数 $= \dfrac{R_p}{R_i}$

$\bar{\nu}/\tau$

平均每一个活性中心单位时间内所消耗的单体分子数

$\dfrac{\bar{\nu}}{\tau}[M\cdot]$

体系中所有活性中心单位时间内所消耗的单体分子数

R_p

$$\bar{\nu} = \frac{R_p}{[M\cdot]}\tau = \frac{R_p}{[M\cdot]}\cdot\frac{[M\cdot]}{R_t} = \frac{R_p}{R_t}$$

$$\bar{\nu} = \frac{R_p}{R_t}$$

瞬时动力学链长

$$\nu = \frac{R_p}{R_i}$$

从统计观点出发

$$\bar{\nu} = \frac{\int_{t_1}^{t_2} R_p \mathrm{d}t}{\int_{t_1}^{t_2} R_t \mathrm{d}t}$$

通过链增长反应消耗的单体数

通过链终止反应消耗的自由基数

- 关系式

$$\nu = \frac{\int_{t_1}^{t_2} R_p \mathrm{d}t}{\int_{t_1}^{t_2} R_t \mathrm{d}t} = \frac{R_p \cdot \Delta t}{R_t \cdot \Delta t} = \frac{R_p}{R_t}$$

时间间隔△t很短

$$\nu = \frac{R_p}{R_i} = \frac{R_p}{R_t} = \frac{k_p[M][M\cdot]}{2k_t[M\cdot]^2} = \frac{k_p[M]}{2k_t[M\cdot]}$$

$$[M\cdot] = \left(\frac{fk_d}{k_t}\right)^{\frac{1}{2}}[I]^{\frac{1}{2}}$$

$$\nu = \frac{k_p^2[M]^2}{2k_t R_p}$$

$$\nu = \frac{k_p[\mathrm{M}]}{2\left(fk_d k_t[\mathrm{I}]\right)^{\frac{1}{2}}} = k_p[M]\tau$$

2. 无链转移的数均聚合度

- 定义

$$\overline{X_n} = \frac{\Delta t\text{时间内加到聚合物上的单体分子数}}{\Delta t\text{时间内生成的聚合物分子数}}$$

  - 累积聚合度

$$\overline{X_n} = \frac{\int_0^t R_p \mathrm{d}t}{\int_0^t R_t \mathrm{d}t}$$

  - 即时聚合度

$$\overline{X_n} = \frac{\int_{t_1}^{t_2} R_p \mathrm{d}t}{\int_{t_1}^{t_2} R_t \mathrm{d}t}$$

- 若t_2-t_1=Δt很短，则R_p，$R_{t,c}$，$R_{t,d}$是常数

$$\overline{X_n}=\frac{\int_{t_1}^{t_2}R_p dt}{\int_{t_1}^{t_2}\left(\frac{R_{t,c}}{2}+R_{t,d}\right)dt}=\frac{R_p\Delta t}{\left(\frac{R_{t,c}}{2}+R_{t,d}\right)\Delta t}=\frac{R_p}{\frac{R_{t,c}}{2}+R_{t,d}}$$

$$=\frac{k_p[\mathrm{M}][\mathrm{M}\cdot]}{\left(k_{t,c}+2k_{t,d}\right)[\mathrm{M}\cdot]^2}=\frac{k_p[\mathrm{M}]}{\left(k_{t,c}+2k_{t,d}\right)\left(\frac{fk_d[\mathrm{I}]}{k_t}\right)^{\frac{1}{2}}}$$

$$=\frac{k_p{}^2[\mathrm{M}]^2}{\left(k_{t,c}+2k_{t,d}\right)R_p}$$

$$\nu=\frac{k_p{}^2[\mathrm{M}]^2}{2k_t\cdot R_p}$$

偶合终止： $\overline{X_n}=2\nu$

歧化终止： $\overline{X_n}=\nu$

两种终止方式共存

$$\overline{X_n} = \frac{-d[M]/dt}{d[P]/dt} = \frac{R_p}{\frac{R_{tc}}{2} + R_{td}}$$

$$C = \frac{k_{tc}}{k_{tc} + k_{td}} \qquad D = \frac{k_{td}}{k_{tc} + k_{td}}$$

$$\overline{X_n} = \frac{1}{\frac{C}{2} + D} \cdot \nu$$

So, as $\nu \propto [M]^1$ & $[I]^{-1/2}$ then $M_n = M \times X_n \propto [M]^1$ & $[I]^{-1/2}$

3.6 自由基聚合反应热力学

1. 基元反应的动力学参数

	$R_d=2fk_d[I]$	$R_p=k_p[M][M\cdot]$	$R_t=2k_t[M\cdot]^2$
E $kJ.mol^{-1}$	105~150	16~34	8~21
$k_{(\ )}$	10^{-4}~$10^{-6}s^{-1}$	10^{2}~$10^{4}L/mol.s^{-1}$	10^{6}~$10^{8}L/mol.s^{-1}$
	f=0.6~0.8 [I]= 10^{-2}~$10^{-4}mol/L$	[M]= 10^{-1}~10mol/L	[M·]= 10^{-7}~$10^{-9}mol/L$
$R_{(\ )}$	10^{-8}~10^{-10}	10^{-4}~10^{-6}	10^{-8}~10^{-10}

2. General Thermodynamics

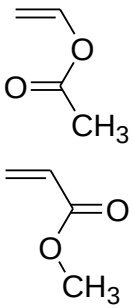
$$\Delta G = \Delta H - T \Delta S$$

- Typical values ΔH at room temperature (298K) for vinyl monomers -40 to -100 kJ mol^{-1}.
- Typically, $\Delta S \sim -100$ J mol^{-1} K^{-1}.

Monomer	ΔH_p (kJ mol^{-1})	ΔS_s (J K^{-1} mol^{-1})
Ethylene	-93	-155
Styrene	-73	-104
Methyl Methacrylate	-56	-117
Tetrafluoroethylene	-163	-112
α-methyl styrene	-35	-110

3.Thermodynamics of Radical Polymerisation

Examples

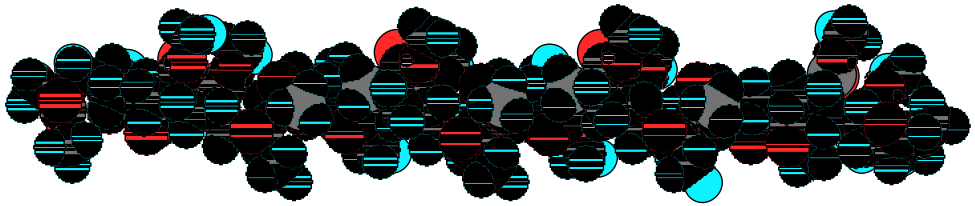


$\Delta H = -77 kJ/mol^{-1}$

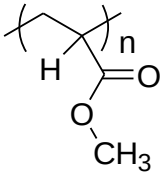
ΔH (MA)

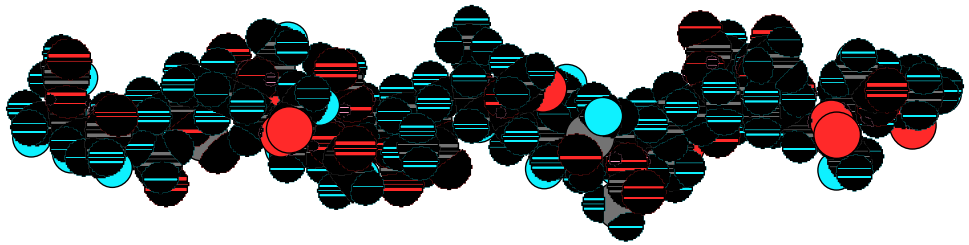
ΔH (VAc)

$\Delta H = -88 kJ/mol^{-1}$

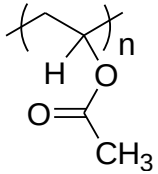


PMA





PVAc



Summary

- Chain Poly of alkenes is exothermic
- Enthalpy of poly (ΔH_p) is negative but entropy of poly (ΔS_p) is also negative.
- Representative values at room temperature (298K) show that free energy of poly is negative

$$\Delta G_p = \Delta H_p - T \Delta S_p$$

Thus as $\Delta H_p > T \Delta S_p$

Then $\Delta G_p < 0$

- Hence polymerisation is thermodynamically feasible

4. 表观活化能

- 总反应速度 R 的表观活化能 E_R

$$R = k_p\left(\frac{fk_d}{k_t}\right)^{\frac{1}{2}}[M][I]^{\frac{1}{2}}$$

$$k_R = k_p\left(\frac{k_d}{k_t}\right)^{\frac{1}{2}} = A_R e^{-\frac{E_R}{RT}} = A_p e^{-\frac{E_p}{RT}}\left(\frac{A_d e^{-\frac{E_d}{RT}}}{A_t e^{-\frac{E_t}{RT}}}\right)^{1/2}$$

$$= A_p \left(\frac{A_d}{A_t} \right)^{\frac{1}{2}} \cdot e^{-(E_p + \frac{1}{2}E_d - \frac{1}{2}E_t)/RT}$$

$$E_R = E_p + \frac{E_d}{2} - \frac{E_t}{2} = 83kJ/mol > 0$$

E_d=125kJ/mol， E_p=29kJ/mol， E_t=17kJ/mol

$T \uparrow$

$E_d \downarrow$

$\longrightarrow R \uparrow$

$E_{光}$=20kJ/mol

■ 动力学链长 ν 的表观活化能 E_ν

$$\nu = \frac{R_p}{R_t} = \frac{k_p[\mathrm{M}]}{2k_t[\mathrm{M}^\bullet]} = \frac{k_p[\mathrm{M}]}{2\left(k_t k_d f[\mathrm{I}]\right)^{1/2}} = \frac{k_p}{2(fk_d k_t)^{\frac{1}{2}}} \cdot \frac{[M]}{[I]^{\frac{1}{2}}}$$

$$\mathrm{k}_\nu = \mathrm{A}e^{-\mathrm{E}/\mathrm{RT}} = \frac{\mathrm{A_P}}{(\mathrm{A_d A_t})^{1/2}} \exp\{-[(\mathrm{E_P} - (\mathrm{E_t}/2 + \mathrm{E_d}/2)]/\mathrm{RT}\}$$

$$E_\nu = E_p - \frac{E_t}{2} - \frac{E_d}{2} < 0$$

若升高温度，则 ν 下降，则 X_n 下降

■ 自由基寿命 τ 的表观活化能 E_τ

$$\tau = \frac{[\mathrm{M}\cdot]}{2fk_d[\mathrm{I}]} = \frac{1}{2\left(fk_d k_t[\mathrm{I}]\right)^{1/2}}$$

$$E_\tau = -\frac{1}{2}E_d - \frac{1}{2}E_t < 0$$

若温度上升，则 τ 下降

$$[M\cdot] = \left(\frac{fk_d}{k_t}\right)^{\frac{1}{2}}[I]^{\frac{1}{2}}$$

5. 上限温度

- Polymerisations have a level of Inherent Safety because they are reversible:

$$M_n^{\bullet} + M \underset{\text{解聚反应：吸热}}{\overset{\text{聚合反应：放热}}{\rightleftharpoons}} M_{n+1}^{\bullet}$$

- Thus we *should* never get to the stage of having an uncontrolled reaction

平衡状态:

$$K = \frac{\left[M_{n+1}^{\bullet}\right]_e}{\left[M_n^{\bullet}\right]_e \left[M\right]_e}$$

$$\left[M_n^{\bullet}\right] \approx \left[M_{n+1}^{\bullet}\right] \longrightarrow K \approx \frac{1}{\left[M\right]_e}$$

上限温度：当温度上升到某个值时，若温度再升高，解聚反应过于剧烈，聚合反应将无法进行，称该临界温度为上限温度 T_c

定义：当$[M]_e$=1mol·L^{-1}时，K=1，此时平衡温度为T_c

5. 上限温度

- At *Ceiling Temp,* T_c , then rate of poly = rate of depoly

- Hence at T_c, $\Delta G_p = 0$ and as $\Delta G_p = \Delta H_p - T\Delta S_p$

平衡状态: $T_c = \Delta H_p / \Delta S_p$

For MMA, $T_c = 198^oC$,

for α -methyl styrene, $T_c = 66^oC$